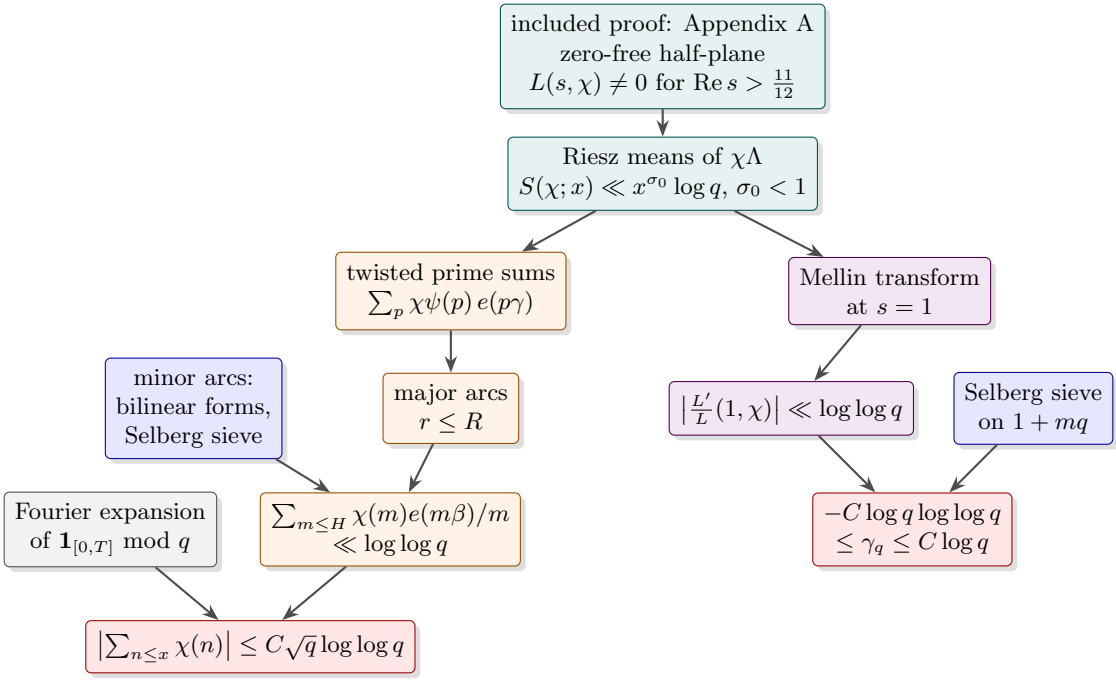

included proof: Appendix A
zero-free half-plane
$L(s,\chi) \neq 0$ for $\operatorname{Re} s > \frac{11}{12}$
Riesz means of $\chi\Lambda$
$S(\chi;x) \ll x^{\sigma_0} \log q$, $\sigma_0 < 1$
twisted prime sums
$\sum_p \chi\psi(p)\, e(p\gamma)$
Mellin transform
at $s = 1$
minor arcs:
bilinear forms,
Selberg sieve
major arcs
$r \leq R$
$\left|\frac{L'}{L}(1,\chi)\right| \ll \log\log q$
Selberg sieve
on $1 + mq$
Fourier expansion
of $\mathbf{1}_{[0,T]}$ mod q
$\sum_{m \leq H} \chi(m) e(m\beta)/m$
$\ll \log\log q$
$-C \log q \log\log q$
$\leq \gamma_q \leq C \log q$
$\left|\sum_{n \leq x} \chi(n)\right| \leq C\sqrt{q} \log\log q$